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MACADAM AGROADVISORY TECHNICAL GUIDE

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SUSTAINABLE IRRIGATION & WATER ADVISORY FIELD NOTES

Soil-Plant-Water Relationships, System Design, and Salinity Management

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CHAPTER 1: SOIL-PLANT-WATER RELATIONSHIPS

1.1 Key Moisture Thresholds

- Saturation: All soil pore space is filled with water; free drainage occurs under gravity. Anaerobic stress begins if sustained > 24 hours.
- Field Capacity (FC): The volume of water retained after gravitational water has drained (typically 1 to 3 days after heavy rainfall/irrigation). Tensiometer reading: 10 - 33 kPa (10 - 33 centibars).
- Permanent Wilting Point (PWP): Soil moisture level where plants can no longer exert sufficient suction pressure to extract water; permanent wilting occurs. Tensiometer reading: ~1,500 kPa.
- Available Water Capacity (AWC): $AWC = FC - PWP$.

1.2 Maximum Allowable Depletion (MAD)

Irrigation must be triggered before soil moisture drops below the MAD threshold to prevent yield loss.

- Standard Field Crops: $MAD = 50\%$ of AWC.
- High-Value/Drought-Sensitive Vegetables: $MAD = 30\% - 40\%$ of AWC.

CHAPTER 2: EVAPOTRANSPIRATION (ET) & SCHEDULING

2.1 Calculating Crop Water Requirements (ET_c)

Crop Evapotranspiration (ET_c) is determined using reference evapotranspiration (ET_o) from local weather station data multiplied by a crop coefficient (K_c):

$$ET_c = ET_o \times K_c$$

Crop Coefficient (K_c) Stages for Row Crops:

1. Initial Stage (Germination/Early vegetative): $K_c \approx 0.3 - 0.4$
2. Mid-Season Stage (Full canopy cover/Reproductive): $K_c \approx 1.05 - 1.20$
3. Late Season Stage (Maturation/Senescence): $K_c \approx 0.4 - 0.6$

2.2 Water Budget Method

Today's Soil Moisture Deficit = Yesterday's Deficit + ΔET_c - Effective Rainfall - Applied Irrigation

CHAPTER 3: SOIL MOISTURE SENSORS & DATA INTERPRETATION

3.1 Sensor Types and Operating Mechanics

- Tensiometers: Measure soil water matric potential (suction). Best suited for high-frequency drip systems in sandy or loamy soils.
- Capacitance / Frequency Domain Reflectometry (FDR) Probes: Measure dielectric permittivity to report Volumetric Water Content (VWC, %).
- Time Domain Reflectometry (TDR): High accuracy across varying soil textures; less sensitive to background salinity changes.

3.2 Interpreting Data Curves

- Stair-Step Pattern: Indicates active daytime root extraction (ΔET_c) and nighttime quiescence.
- Flat Line at Top: Indicates deep percolation loss below the sensor zone.
- Sharp Decline: Indicates root extraction is moving down to deeper soil horizons as shallow moisture depletes.

CHAPTER 4: IRRIGATION SYSTEMS: UNIFORMITY & ENGINEERING

4.1 Catch-Can Distribution Uniformity (DU) Test

Distribution Uniformity measures how evenly water is applied across the field:

$$DU = \frac{\text{Average depth of lowest quarter of catch cans}}{\text{Average depth of all catch cans}} \times 100$$

- Excellent DU Rating: > 85%
- Acceptable DU Rating: 75% - 85%
- Action Required (System Maintenance Needed): < 70%

4.2 System Maintenance Checklist

1. Pressure Regulators: Check pressure drop across drip zones or pivot spans.
2. Filtration Systems: Backwash media filters when pressure differential (ΔP) exceeds 5 - 7 PSI.
3. Flushing Protocols: Regularly open lateral ends to flush sediment accumulation.

CHAPTER 5: WATER QUALITY, SALINITY & LEACHING FRACTIONS

5.1 Key Water Quality Indicators

- Electrical Conductivity of Water (EC_w): Measures total dissolved salts. Target < 1.0 dS/m.
- Sodium Adsorption Ratio (SAR): Measures sodium hazard relative to calcium and magnesium:

$$SAR = \frac{[Na^+]}{\sqrt{\frac{[Ca^{2+}] + [Mg^{2+}]}{2}}}$$

Target SAR < 6.0 to prevent soil permeability breakdown.

5.2 Calculating Leaching Requirement (LR)

To prevent salt build-up in the root zone when using saline water:

$$LR = \frac{EC_w}{5 \times EC_e - EC_w}$$

Where EC_e is the target soil salinity threshold for the specific crop.

CHAPTER 6: DROUGHT MITIGATION & CONSERVATION TILLAGE

- Mulching: Organic mulch reduces soil surface evaporation losses by 30% - 50%.
- Crop Residue Management: Maintaining > 30% surface cover increases water infiltration rates and reduces surface crusting during high-intensity rain events.